```
TITLE 'Soft Drink Bottling Experiment Example 5.3';
DATA BOTTLING;
INPUT PRESSURE SPEED CARBONATION REPLICATE FILLHEIGHT @@;
      CARDS;
25 200 10 1 -3  25 200 10 2 -1  25 200 12 1 0 25 200 12 2 1  25 200 14 1  5  25 200 14 2  4
25 250 10 1 -1  25 250 10 2  0  25 250 12 1 2 25 250 12 2 1  25 250 14 1  7  25 250 14 2  6
30 200 10 1 -1  30 200 10 2  0  30 200 12 1 2 30 200 12 2 3  30 200 14 1  7  30 200 14 2  9
30 250 10 1  1  30 250 10 2  1  30 250 12 1 6 30 250 12 2 5  30 250 14 1 10  30 250 14 2 11

;
PROC GLM PLOTS=(DIAGNOSTICS);
     CLASS PRESSURE SPEED CARBONATION REPLICATE;
          MODEL FILLHEIGHT=PRESSURE|SPEED|CARBONATION;
               MEANS PRESSURE|SPEED|CARBONATION/TUKEY LINES;
                     OUTPUT OUT=NEW P=PREDY R=RESID;
PROC UNIVARIATE NORMAL;
VAR RESID;
RUN;
```